WORKED SOLUTION

STEP III 2015, Question 3

Pure • Polar Curves and Area

STEP III 2015, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q3

The key idea

The curve $r = a\sec\theta$ is just the straight line $x = a$, because $r\cos\theta = x$. So $r - a\sec\theta$ measures, along the ray from the origin at angle θ, how far a point is from where that ray meets the line, and the equation says that this distance is b. That is exactly the definition of a conchoid, with the pole as the fixed point. For the sketches, unpick the modulus into $r = a\sec\theta + b$ and $r = a\sec\theta - b$ and ask, for each, which values of θ give $r \geq 0$.

Solution

Throughout, $\sec\theta$ is undefined at $\theta = \pm\frac{\pi}{2}$, so those angles are excluded. Removing the modulus, $(*)$ says

$$r = a\sec\theta + b \qquad \text{or} \qquad r = a\sec\theta - b,$$

and in either case we also need $r \geq 0$.

Part (i)

Why $\sec\theta > 0$. Suppose $\sec\theta < 0$. Then $|\cos\theta| \leq 1$ gives $\sec\theta \leq -1$, so $a\sec\theta \leq -a$. Both possible values of r then satisfy

$$r = a\sec\theta \pm b \leq -a + b < 0,$$

using $a > b$. This contradicts $r \geq 0$, so every point satisfying $(*)$ has $\sec\theta > 0$, that is, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

The conchoid. Take A to be the pole O, L to be the line $x = a$ (in polar form $r\cos\theta = a$, or $r = a\sec\theta$), and $d = b$. Note that A is not on L, since $a > 0$.

Let P satisfy $(*)$, with polar angle θ, so $\cos\theta > 0$. The line OP is not parallel to L, and it meets L at the point Q on the same ray with $OQ = a\sec\theta$, i.e. $Q = (a, a\tan\theta)$ in Cartesians. Since O, Q and P all lie on one ray from O, at distances $a\sec\theta$ and r from

O,

$$PQ = |r - a\sec\theta| = b.$$

So P is at distance b from L, measured along a line through $A = O$: it lies on the conchoid of Nicomedes with this L, A and $d = b$.

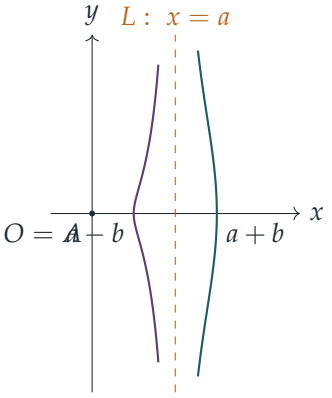
(Conversely, every point of that conchoid satisfies $(*)$: on any line through O meeting L at Q, the two points at distance b from Q are at distances $a\sec\theta \pm b$ from O along the ray OQ, both positive because $a\sec\theta \geq a > b$. So the locus is the whole conchoid.)

Sketch. In Cartesian coordinates, $x = r\cos\theta = a \pm b\cos\theta$ and $y = r\sin\theta$. Multiplying $r - a\sec\theta = \pm b$ by $\cos\theta$ gives $x - a = \pm b\cos\theta = \pm bx/r$, so the curve is

$$(x-a)^2(x^2+y^2) = b^2x^2.$$

There are two branches, both symmetric about the x-axis (replace θ by $-\theta$) and both asymptotic to L as $\theta \to \pm\frac{\pi}{2}$, where $r \to \infty$:

- the outer branch $r = a\sec\theta + b$, with $x = a + b\cos\theta \in (a,\, a+b]$, crossing the axis at $(a+b, 0)$;
- the inner branch $r = a\sec\theta - b$, with $x = a - b\cos\theta \in [a-b,\, a)$, crossing the axis at $(a-b, 0)$, which is to the right of O.



(The picture uses $a = 2$, $b = 1$: the outer branch in teal, the inner in plum.)

Part (ii)

Now $a < b$. Go through the two cases again, checking where $r \geq 0$.

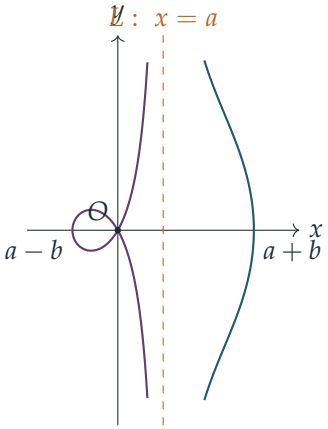
- $r = a\sec\theta + b$. If $\sec\theta > 0$ this is always positive: the same outer branch as before, to the right of L. If $\sec\theta < 0$, then $r \geq 0$ exactly when $-a\sec\theta \leq b$, i.e. $|\sec\theta| \leq b/a$, i.e. $\cos\theta \leq -a/b$.
- $r = a\sec\theta - b$. If $\sec\theta < 0$ this is negative, so there are no points. If $\sec\theta > 0$, then $r \geq 0$ exactly when $\sec\theta \geq b/a$, i.e. $0 < \cos\theta \leq a/b$.

Let $\alpha = \arccos(a/b)$, which lies in $(0, \frac{\pi}{2})$ because $0 < a/b < 1$.

Ask yourself What does the piece with $\sec\theta < 0$ look like?

It is $r = b + a\sec\theta$ for $\cos\theta \le -a/b$, that is, $\pi - \alpha \le |\theta| \le \pi$. At the ends, $\cos\theta = -a/b$ gives $r = b - b = 0$, so it starts and finishes at the origin. At $\theta = \pi$ we get $r = b - a$, the point $(a-b, 0)$ on the negative x-axis. In between $r > 0$, and the curve is symmetric about the x-axis, so this piece is a closed loop to the left of the origin, lying in $a - b \le x \le 0$ (since $x = a + b\cos\theta$).

The inner branch $r = a\sec\theta - b$, for $\alpha \le |\theta| < \frac{\pi}{2}$, also starts at the origin (where $\cos\theta = a/b$) and runs out to the asymptote L as $|\theta| \to \frac{\pi}{2}$, staying in $0 \le x < a$. So the inner branch and the loop together form one curve which crosses itself at O, with tangent directions $\theta = \pm\alpha$ there.



(The picture uses $a = 1$, $b = 2$, so $\alpha = \frac{\pi}{3}$. The outer branch is in teal; the inner branch together with the loop is in plum.)

Area of the loop for $a = 1$, $b = 2$. Here $\cos\theta \le -\frac{1}{2}$ means $\frac{2\pi}{3} \le |\theta| \le \pi$, and on the loop $r = 2 + \sec\theta$. As θ runs through this range the ray sweeps the loop exactly once, so the area is $\frac{1}{2}\int r^2\,\mathrm{d}\theta$. Using the symmetry about the x-axis, we take twice the upper half, $\frac{2\pi}{3} \le \theta \le \pi$:

$$\begin{aligned}\text{Area} &= 2\cdot\frac{1}{2}\int_{2\pi/3}^{\pi}(2+\sec\theta)^2\,\mathrm{d}\theta = \int_{2\pi/3}^{\pi}\left(4 + 4\sec\theta + \sec^2\theta\right)\mathrm{d}\theta \\ &= \Big[\,4\theta + 4\ln|\sec\theta + \tan\theta| + \tan\theta\,\Big]_{2\pi/3}^{\pi}.\end{aligned}$$

The given antiderivative of $\sec\theta$ is valid here because the interval does not contain $\frac{\pi}{2}$. At $\theta = \pi$: $\sec\theta = -1$, $\tan\theta = 0$, so the bracket is $4\pi + 4\ln 1 + 0 = 4\pi$. At $\theta = \frac{2\pi}{3}$: $\sec\theta = -2$, $\tan\theta = -\sqrt{3}$, so the bracket is $\frac{8\pi}{3} + 4\ln(2+\sqrt{3}) - \sqrt{3}$. Subtracting,

$$\text{Area of loop} = \frac{4\pi}{3} + \sqrt{3} - 4\ln\left(2+\sqrt{3}\right).$$

Checks

- Numerically the area is $4.1888 + 1.7321 - 5.2678 \approx 0.6530$. It is positive, as it must be, and small compared with the box $-1 \le x \le 0$, $|y| \le 0.45$ that the loop fits in (whose area is 0.9). A direct numerical integration of $\frac{1}{2}(2+\sec\theta)^2$ over $[\frac{2\pi}{3}, \frac{4\pi}{3}]$ gives 0.65301 as well.
- The loop passes through O and $(-1,0)$: at $\theta = \frac{2\pi}{3}$, $r = 2 - 2 = 0$; at $\theta = \pi$, $r = 2 - 1 = 1$.
- Pick a point on the curve in (i), say $a = 2$, $b = 1$, $\theta = \frac{\pi}{4}$: then $r = 2\sqrt{2} + 1$ and the ray meets L at $(2,2)$, distance $2\sqrt{2}$ from O, so the point is $1 = b$ beyond it.

Common slips

- Forgetting the condition $r \ge 0$. It is what forces $\sec\theta > 0$ in (i), and it fixes the range of θ for the loop in (ii).
- Integrating the loop over $\frac{\pi}{2} < \theta < \pi$, through the asymptote direction, instead of starting where $r = 0$ at $\theta = \frac{2\pi}{3}$.
- Sign errors with $\ln|\sec\theta + \tan\theta|$ in the second quadrant, where both $\sec\theta$ and $\tan\theta$ are negative.

Beyond STEP

The conchoid was invented by Nicomedes around 200 BC for trisecting angles and finding cube roots, problems that are impossible with straight edge and compass alone; the impossibility proofs come from field theory, met in a first course in algebra (see Allenby, *Rings, Fields and Groups*). The self-crossing at the origin in (ii) is a *node*, an example of a singular point on an algebraic curve, here the quartic $(x-a)^2(x^2+y^2) = b^2x^2$. For many more STEP-style curve-sketching problems, see Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.